

Smart Home Lighting Automation System

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TASK -3

Robotics & Automation — Internship

1. Introduction

Modern residential and commercial spaces increasingly demand intelligent, energy-efficient lighting solutions. A

Smart Home Lighting Automation System integrates sensors, microcontrollers, and actuators to control artificial lighting automatically, eliminating the need for manual switching while optimising power consumption.

This report presents the complete design and working principle of such a system, covering hardware components, circuit operation, control logic, and real-world applications. The system is built around an Arduino-based microcontroller that reads ambient light and occupancy data to make real-time lighting decisions.

The system monitors two environmental parameters — **ambient light intensity** and **human presence** — and drives LED loads through a relay interface. A Wi-Fi module optionally extends control to a smartphone application, enabling remote override and scheduling.

Key design objectives:

- Automatic ON/OFF control based on occupancy and daylight availability.
- Energy saving by preventing lights from remaining ON in unoccupied rooms.
- Manual override capability via physical switch or mobile app.
- Scalable architecture — additional rooms/zones can be added modularly.
- Low-cost, open-source implementation using widely available components.

3. Components Used

Component	Specification / Model	Function
Microcontroller	Arduino Uno (ATmega328P)	Central processing & decision logic
Light Sensor (LDR)	GL5528 Photoresistor	Measures ambient light intensity

PIR Motion Sensor	HC-SR501	Detects human presence / motion
Relay Module	5 V, 1-channel / 4-channel	Switches mains-voltage LED loads
LED Light / Bulb	12 V DC LED strip or AC bulb	Output lighting load
Wi-Fi Module	ESP8266 (NodeMCU)	IoT connectivity for remote control
Power Supply	5 V / 2 A DC adapter	Powers microcontroller & sensors
Resistors	10 kΩ pull-down resistors	Signal conditioning for LDR
Connecting Wires	Jumper wires / PCB traces	Electrical interconnections
Enclosure / PCB	Prototype board or custom PCB	Mechanical housing

4. Working Principle

The system operates on a dual-condition logic: the light is activated **only** when both (a) the ambient light level falls below a preset threshold AND (b) a person is detected in the room. This prevents unnecessary illumination even during nighttime when the room is unoccupied.

The LDR and a fixed resistor form a voltage divider. The Arduino reads the analog voltage at the midpoint — lower voltage indicates higher light (LDR resistance drops). The PIR sensor outputs a digital HIGH pulse (typically lasting 3–5 seconds, adjustable) when infrared radiation from a moving warm body is detected. The microcontroller evaluates both inputs at every loop cycle (~100 ms) and sets the relay output accordingly.

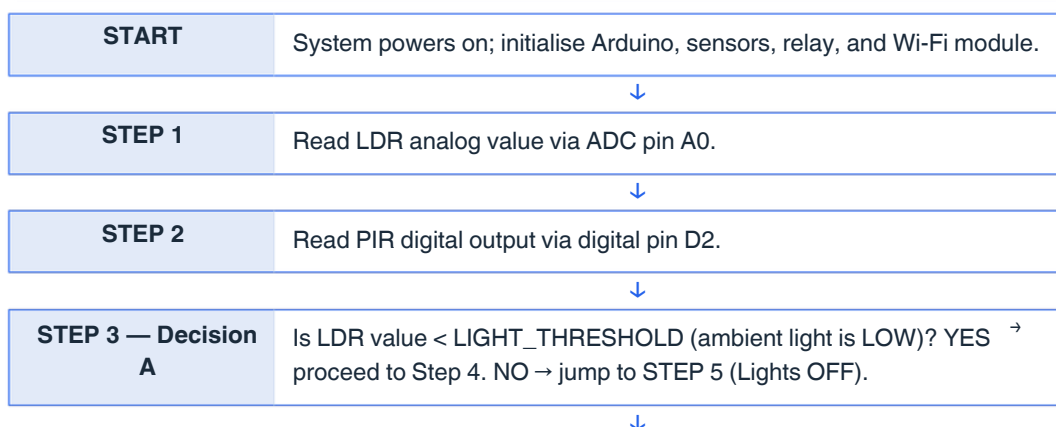
Control Logic (pseudocode):

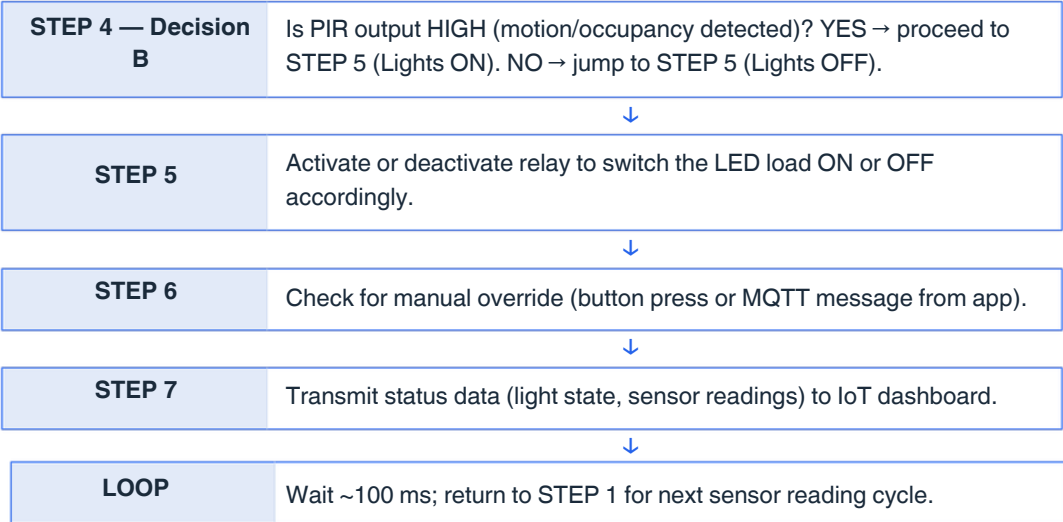
```
IF (ldr_value < LIGHT_THRESHOLD) AND (pir_state == HIGH):
    relay_output = ON // Lights ON
ELSE:
    relay_output = OFF // Lights OFF

//LIGHT_THRESHOLD is calibrated during setup (typical: 400 on 0–1023 ADC scale)
```

The Wi-Fi module subscribes to an MQTT broker, allowing a smartphone app to publish override commands. A manual push-button ISR (Interrupt Service Routine) provides hardware-level override independent of software state.

5. System Flowchart





6. Circuit Description

The circuit is logically divided into three sub-blocks: **Sensing**, **Processing**, and **Actuation**.

6.1 Sensing Sub-circuit

The LDR is connected in series with a 10 kΩ resistor between VCC (5 V) and GND. The junction node feeds Arduino analogue pin A0. In darkness, LDR resistance is high (~1 MΩ), pulling A0 toward 5 V (ADC reads ~900–1023). Under bright light, resistance drops (~200 Ω), pulling A0 toward 0 V (ADC reads ~0–100).

The HC-SR501 PIR sensor is self-contained with built-in Fresnel lens and amplifier. Its VCC and GND connect to the 5 V rail; its OUT pin connects directly to Arduino digital pin D2, which is configured with an internal pull-down and external interrupt.

6.2 Processing Sub-circuit

The Arduino Uno serves as the control core. It requires no external clock components beyond the onboard 16 MHz crystal. The ESP8266 module communicates with the Arduino via UART (TX/RX on pins D10/D11 using SoftwareSerial) at 9600 baud.

6.3 Actuation Sub-circuit

The relay module's IN pin connects to Arduino digital pin D7. When D7 goes HIGH, the relay coil energises, closing the Normally Open (NO) contact and completing the AC/DC mains circuit to the LED load. A freewheeling diode (1N4007) across the relay coil suppresses back-EMF. The relay electrically isolates the low-voltage control circuit from the high-voltage load circuit, ensuring safety.

7. Advantages

- Energy Efficiency:** Lights operate only when needed, reducing electricity consumption by up to 40–60% compared to manual switching.
- Convenience:** Fully hands-free operation; no physical switching required by occupants.
- Cost-Effective:** Built with low-cost, readily available open-source hardware (total BOM cost ≈ RS 500–RS 1,500).
- Scalability:** Modular zone design allows easy expansion to multiple rooms.
- Remote Control:** Wi-Fi integration enables monitoring and control from anywhere via smartphone.
- **Safety:** Relay isolation protects low-voltage electronics from mains-voltage loads.
- **Adaptability:** Threshold values are software-configurable without hardware changes.

8. Applications

Domain	Application
Residential	Bedrooms, bathrooms, stairwells, garages, outdoor porch lighting
Commercial	Office corridors, conference rooms, restrooms, parking lots
Hospitality	Hotel room automation, lobby and hallway lighting management
Healthcare	Hospital ward lighting, automated night-mode in patient rooms
Industrial	Warehouse aisle lighting, machine-area safety lighting
Smart Cities	Street light automation based on ambient light and traffic sensors
Educational	Classroom and laboratory automation for energy conservation

9. Conclusion

TheSmartHome Lighting Automation System demonstrates a practical and efficient application of embedded systems and IoT technology in everyday environments. By combining an LDR, a PIR sensor, an Arduino microcontroller, and a relay-driven output stage, the system achieves intelligent, automatic lighting control that conserves energy without sacrificing user comfort.

The dual-condition control logic ensures lights are ON only when both the ambient illumination is insufficient AND a human occupant is present — a simple yet powerful design decision that maximises efficiency. The optional Wi-Fi connectivity layer elevates the system from a standalone device to a fully networked IoT node, compatible with broader smart-home ecosystems.

Future enhancements may include integration with voice assistants (Google Home / Alexa), machine learning-based occupancy prediction, daylight harvesting algorithms, and energy metering for real-time consumption analytics. This project serves as a strong foundation for more advanced building automation systems.